

Killed Noncolliding Walkers on a Finite Interval: Determinantal Spectrum, Absorption, and the Q-Process

Route-A source-local certificate HCS-C306

3 September 2026

Abstract

For every $1 \leq k \leq L$, we diagonalize k continuous-time symmetric nearest-neighbour walks on $\{1, \dots, L\}$ killed at the first boundary hit or collision. The one-particle sine kernel and the Karlin–McGregor determinant yield a complete Slater basis of all $\binom{L}{k}$ chamber states.

1 Killed chamber dynamics

Fix integers $L \geq 1$ and $1 \leq k \leq L$. A single particle jumps one site left and right at rate one, and is killed on an attempted jump to 0 or $L + 1$. The k particles have independent clocks until the first such boundary attempt or the first attempted coincidence. Before killing their ordered configuration lies in

$$\mathcal{W}_{L,k} = \{x = (x_1, \dots, x_k) : 1 \leq x_1 < \dots < x_k \leq L\}.$$

For f on this chamber, extended by zero across collision and boundary faces, the sub-Markov generator is

$$(Q_k f)(x) = \sum_{a=1}^k \{f(x - e_a) + f(x + e_a) - 2f(x)\}. \quad (1)$$

Thus every legal one-coordinate step has rate one and $Q_k(x, x) = -2k$. This diagonal is essential: a forbidden attempt kills; it is not suppressed or reflected as in an exclusion chain.

For one particle put

$$\phi_r(j) = \sqrt{\frac{2}{L+1}} \sin \frac{\pi r j}{L+1}, \quad \varepsilon_r = 2 - 2 \cos \frac{\pi r}{L+1}, \quad 1 \leq r, j \leq L. \quad (2)$$

The Dirichlet difference equation and sine orthogonality give

$$p_t^{(1)}(i, j) = \frac{2}{L+1} \sum_{r=1}^L e^{-\varepsilon_r t} \sin \frac{\pi r i}{L+1} \sin \frac{\pi r j}{L+1}. \quad (3)$$

2 Determinant and complete spectrum

For increasing mode indices $m = (m_1 < \dots < m_k)$ define

$$\Phi_m(x) = \det[\phi_{m_a}(x_b)]_{a,b=1}^k, \quad \Lambda_m = \sum_{a=1}^k \varepsilon_{m_a}. \quad (4)$$

Theorem 1 (Full killed semigroup atlas). *The $\binom{L}{k}$ functions Φ_m form a complete orthonormal basis of $\ell^2(\mathcal{W}_{L,k})$ and $Q_k \Phi_m = -\Lambda_m \Phi_m$. For all $t \geq 0$,*

$$P_t(x, y) = \Pr_x\{X(t) = y, \tau > t\} = \det[p_t^{(1)}(x_i, y_j)]_{i,j=1}^k, \quad (5)$$

$$P_t(x, y) = \sum_m e^{-\Lambda_m t} \Phi_m(x) \Phi_m(y). \quad (6)$$

Every eigenvalue is displayed with its multiplicity, including coincidences among distinct mode sums.

Proof. Tensor products of the one-particle eigenvectors diagonalize the generator of the independent product chain. Antisymmetrizing a tensor product gives the determinant in (4); crossing a wall changes sign, so its restriction to the ordered chamber obeys the zero boundary convention in (1). Cauchy–Binet, or direct antisymmetric tensor orthogonality, gives $\sum_{x \in \mathcal{W}_{L,k}} \Phi_m(x) \Phi_n(x) = \mathbf{1}_{m=n}$. There are exactly $\binom{L}{k} = |\mathcal{W}_{L,k}|$ modes, proving completeness and (6).

For (5), expand the determinant of (3) and apply Cauchy–Binet. Equivalently, the Karlin–McGregor path-switching involution pairs product paths at their first collision with opposite permutation sign; only never-coincident paths remain. Boundary killing is already contained in each one-particle kernel. At $t = 0$, orthonormal completeness makes either formula the identity kernel. $\square$

The lowest mode is $m_0 = (1, \dots, k)$. Its determinant has a fixed sign. Indeed, with $\theta_b = \pi x_b / (L + 1)$ and $\sin(a\theta) = \sin \theta U_{a-1}(\cos \theta)$,

$$h(x) := (-1)^{k(k-1)/2} \Phi_{m_0}(x) = \left(\frac{2}{L+1} \right)^{k/2} 2^{k(k-1)/2} \prod_b \sin \theta_b \prod_{i < j} (\cos \theta_i - \cos \theta_j) > 0. \quad (7)$$

Thus $\|h\|_2 = 1$ and $\Lambda_0 = \sum_{r=1}^k \varepsilon_r$ is simple.

Source ownership and AI-use statement

Karlin and McGregor [1] established the coincidence determinant for birth–death processes. Darroch and Seneta [2] treat quasi-stationarity in finite continuous-time absorbing chains. We claim no priority for those results; the contribution here is a closed source-local derivation and auditable finite certificate for the frozen interval model. AI tools assisted algebra checking, hostile mutation design, and manuscript preparation. Every theorem dependency, numerical cutoff, and nonclaim is exposed in the package.

References

- [1] S. Karlin and J. McGregor, “Coincidence probabilities,” *Pacific Journal of Mathematics* 9 (1959), 1141–1164, doi:10.2140/pjm.1959.9.1141.
- [2] J. N. Darroch and E. Seneta, “On quasi-stationary distributions in absorbing continuous-time finite Markov chains,” *Journal of Applied Probability* 4 (1967), 192–196, doi:10.2307/3212311.